Given a map $f: \mathbb{R} \to \mathbb{R}$, a map

$$g: x \mapsto \int_J K(x,t) f(t)\, dt$$

is called the integral transform.

In this page, we cover typical int. forms.

「Laplace transform.

Given a map $f: \mathbb{R} \to \mathbb{C}$, a Laplace transform of f is given by:

$$\mathcal{L}f : \mathbb{C} \to \mathbb{C} ; x \mapsto \int_0^\infty e^{-xt} \cdot f(t)\, dt.$$

「Prop. If f is differentiable and $e^{-x \cdot R} \cdot f(R) \to 0$ as $R \to \infty$,

then $$(\mathcal{L}f')(x) = \int_0^\infty e^{-xt} \cdot f'(t)\, dt = \lim_{R \to \infty} \left[e^{-xt} f(t) \Big|_0^R \right] + x \cdot \int_0^\infty e^{-xt} \cdot f(t)\, dt$$

$$= x \cdot (\mathcal{L}f)(x) - f(0)$$

「Fourier transform.

「Convolution

Given two maps $f, g : \mathbb{R} \to \mathbb{C}$, define

$$(f * g)(x) = \int_{-\infty}^{\infty} g(x-t) f(t)\, dt \qquad x - t \in [-r, r] \Leftrightarrow -r + x$$

「Prop. Let $k_r(x) = \frac{1}{2r} \cdot \chi_{[-r,r]}(x)$. Then,

$$(f * k_r)(x) = \int_{-\infty}^{\infty} k_r(x-t) \cdot f(t)\, dt = \frac{1}{2r} \cdot \int_{x-r}^{x+r} f(t)\, dt \qquad (1)$$

Moreover, $\int_{-\infty}^{\infty} |f| < \infty \Rightarrow f * k_r$ is uniformly continuous. (2)

and f is uniformly continuous $\Rightarrow f * k_r \to f$ as $r \to 0$ uniformly. (3)

Proof. (1) Can be shown by:

$$g(x-t) \cdot f(t) = 0 \Leftrightarrow x - t \in [-r, r] \Leftrightarrow x - r \leq t \leq x + r.$$

To show (2), let $r > 0$ be fixed. Given $\varepsilon > 0$, take $\delta_0 > 0$ such that

$$0 < t - s < \delta_0 \Rightarrow \int_s^t |f| < r \cdot \varepsilon.$$

Now, put $\delta = \min\{2r, \delta_0\}$. Then, $|x - y| < \delta$ implies

$$|(f * k_r)(x) - (f * k_r)(y)| = \frac{1}{2r} \cdot \left| \int_{x-r}^{x+r} f(t)\, dt - \int_{y-r}^{y+r} f(t)\, dt \right| \qquad |x - y| < \delta < 2r.$$

$$= \frac{1}{2r} \cdot \left| \int_{x-r}^{y-r} f(t)\, dt - \int_{x+r}^{y+r} f(t)\, dt \right|$$

$$< \frac{1}{2r} \cdot (r\varepsilon + r\varepsilon) = \varepsilon.$$

To show (3), let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ such that

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Now, for $r < \delta$,

$$|f(x) - (f * k_r)(x)| = \left| \frac{1}{2r} \cdot \int_{x-r}^{x+r} f(x)\, dt - \frac{1}{2r} \cdot \int_{x-r}^{x+r} f(t)\, dt \right|$$

$$\leq \frac{1}{2r} \cdot \int_{x-r}^{x+r} |f(x) - f(t)|\, dt < \varepsilon.$$

Therefore, $f * k_r \to f$ uniformly. ☒